As a tool in working out the star's structure, Chandrasekhar needed to know the answer to the following question: Suppose that white-dwarf matter has already been compressed to some density (for example, a density of a million grams per cubic centimeter). Compress the matter (that is, reduce its volume and increase its density) by an additional 1 percent. The matter will protest against this additional compression by raising its pressure. By what percentage will its pressure go up? Physicists use the name *adiabatic index* for the percentage increase in pressure that results from a 1 percent additional compression. In this book I shall use the more graphic name *resistance to compression* or simply *resistance.* (This "resistance to compression" should not be confused with "electrical resistance"; they are completely different concepts.)

Chandrasekhar worked out the resistance to compression by examining step by step the consequences of a 1 percent increase in the density of white-dwarf matter: the resulting decrease in electron cell size, the decrease in electron wavelength, the increase in electron energy and speed, and finally the increase in pressure. The result was clear: A 1 percent increase in density produced ⅝ of a percent (1.667 percent) increase in pressure. The resistance of white-dwarf matter, therefore, was ⅝.

Many decades before Chandrasekhar's voyage, astrophysicists had computed the details of the balance of gravity and pressure inside any star whose matter has a resistance to compression that is independent of depth in the star-that is, a star whose pressure and density increase in step with each other, as one moves deeper and deeper into the star, with a 1 percent increase in density always accompanied by the same fixed percentage increase in pressure. The details of the resulting stellar structures were contained in Eddington's book *The Internal Constitution of the Stars,* which Chandrasekhar had brought on board the ship because he treasured it so much. Thus, when Chandrasekhar discovered that white-dwarf matter has a resistance to compression of ⅝, independent of its density, he was pleased. He could now go directly to Eddington's book to discover the star's internal structure: the manner in which its density and pressure changed from surface to center.

Among the things that Chandrasekhar discovered, by combining the formulas in Eddington's book with his own formulas, were the density at the center of Sirius B, 360,000 grams per cubic centimeter (6 tons per cubic